Chemical Names and Formulas

Ionic and molecular compounds

Name _____ Class _____ Date _____

IONS

Cations: H^+ · Na^+ · K^+ · Ag^+ · NH_4^+ · Mg^{2+} · Ca^{2+} · Ba^{2+} · Zn^{2+} · Al^{3+} · Fe^{2+}/Fe^{3+} · Cu^+/Cu^{2+}

Anions: F^- · Cl^- · Br^- · I^- · OH^- · NO_3^- · HCO_3^- · O^{2-} · S^{2-} · SO_4^{2-} · CO_3^{2-} · PO_4^{3-} · N^{3-}

Prefixes: mono 1 · di 2 · tri 3 · tetra 4 · penta 5 · hexa 6 · hepta 7 · octa 8 · nona 9 · deca 10

For each ionic formula, show that the charges add up to zero.

A. Ionic or molecular?

1. Classify each compound as ionic or molecular: (a) KBr (b) SO_2 (c) NH_4Cl (d) CCl_4 (e) Fe_2O_3

B. Ionic compounds

2. Write the formula of: (a) potassium oxide (b) magnesium nitride (c) zinc chloride (d) barium sulfate

3. Write the formula of: (a) calcium nitrate (b) aluminium hydroxide (c) ammonium carbonate (d) sodium hydrogencarbonate

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4. Name: (a) Na_2S (b) Na_2SO_4 (c) $AgNO_3$ (d) $Mg_3(PO_4)_2$

5. Name each compound, using a Roman numeral, and show how you found the charge: (a) $FeCl_2$ (b) Fe_2O_3 (c) Cu_2O (d) $CuSO_4$

C. Molecular compounds

6. Name: (a) CO_2 (b) PCl_5 (c) N_2O (d) SF_6. Then write the formula of: (e) sulfur trioxide (f) dinitrogen tetroxide

D. Acids

7. Write the formula of: (a) hydrochloric acid (b) nitric acid (c) sulfuric acid (d) phosphoric acid

E. Challenge

8. A student writes “copper sulfate is $Cu(SO_4)_2$” and “dicalcium dichloride is $CaCl_2$”. Explain each error and give the correct formula or name.

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